MIE 312 Tutorial 4 Summary
Ch.2 – Buoyancy & Pressure on Curved Surfaces

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Learning Objectives

- Understand the concept of buoyancy.
- Calculate buoyancy forces on fully and partially submerged objects.
- Determine floating equilibrium and stability.
- Calculate hydrostatic forces on curved surfaces.

Interactive Tools

- Buoyancy: https://howardho-cfd.github.io/teaching/fluids/interactive-tools/buoyancy/

Buoyancy

Archimedes' Principle

- A submerged object experiences pressure on all surfaces.
- Since pressure increases with depth, the pressure on the bottom is greater than on the top.
- The resultant upward force is the **buoyant force**.

$$F_B = F_{lower} - F_{upper}$$

$$\boxed{F_B = \rho_f g V_{sub}}$$

  where

  - ρ_f = fluid density
  - V_{sub} = submerged volume

- Atmospheric pressure cancels out when acting on both sides of the object.
- The buoyant force acts through the **Centre of Buoyancy (CB)**, which is the centroid of the displaced fluid volume.

Floating Objects

- For a floating object in equilibrium,

$$F_B = W$$

$$\rho_f g V_{sub} = \rho_{obj} g V_{total}$$

- Therefore,

$$\boxed{\frac{V_{sub}}{V_{total}} = \frac{\rho_{obj}}{\rho_f}}$$

- A less dense object has a smaller submerged fraction.
- Example: Ice has $SG \approx 0.92$, so approximately 92% of its volume is submerged.

Partially Submerged Objects

- If the upper surface is exposed to air,

$$P_{upper} \approx P_{atm}$$

- The buoyancy equation remains

$$F_B = \rho_f g V_{sub}$$

- Only the submerged volume contributes to buoyancy.
- As the submerged volume changes, the Centre of Buoyancy (CB) moves.
- The object's weight remains constant (unless its mass changes).

Floating Stability

Key Points

- CG = Centre of Gravity
- CB = Centre of Buoyancy
- M = Metacentre
- Stability depends on the relative positions of CG, CB and M.

Metacentric Height

$$\boxed{MG = \frac{I_0}{V_{sub}} - GB}$$

where

- I_0 = second moment of area of the waterline (top view)
- V_{sub} = submerged volume
- GB = distance between the centre of gravity and centre of buoyancy

Interpretation:

$$MG > 0 \quad \Rightarrow \quad \text{Stable}$$

$$MG = 0 \quad \Rightarrow \quad \text{Neutral Stability}$$

$$MG < 0 \quad \Rightarrow \quad \text{Unstable}$$

Pressure on Curved Surfaces

Basic Idea

- Hydrostatic pressure always acts normal to the surface.
- Resolve the resultant force into horizontal and vertical components.

$$F_R = \sqrt{F_H^2 + F_V^2}$$

$$\theta = \tan^{-1}\left(\frac{F_V}{F_H}\right)$$

Horizontal Component

- Treat the curved surface as its **vertical projection**.
- Calculate pressure exactly as for a plane surface.

$$\boxed{F_H = \gamma h_{CG} A_v}$$

  where

  - A_v = vertical projected area
  - h_{CG} = centroid depth of the projected area

- The line of action passes through the Centre of Pressure of the projected area.

Vertical Component

- Imagine the volume of fluid directly above the curved surface.
- The vertical force equals the weight of this imaginary fluid.

$$\boxed{F_V = \gamma V}$$

- The line of action passes through the centroid of the imaginary fluid volume.
- F_V acts downward if the fluid is above the surface and upward if the fluid is below the surface.

Problem-Solving Procedure

1. Draw the curved surface and the fluid volume.
2. Calculate the horizontal force using the vertical projection.
3. Calculate the vertical force using the fluid volume.
4. Determine the line of action of each component.
5. Combine the components to obtain the resultant force and its direction.
6. Apply equilibrium equations if moments or support reactions are required.